

EXP.NO: 7	Producer Consumer
DATE:	

AIM:

To implement the Producer-Consumer problem using semaphores, ensuring mutual exclusion and preventing buffer overflow (Full) or underflow (Empty) during process execution.

ALGORITHM:

- **Initialize Controls:** Start with a lock (**m=1**), 5 empty slots (**e=5**), and 0 full slots (**f=0**).
- **Producer Step:**
 - Check if space is available.
 - Lock the system, fill one empty slot, increase the full count, and then unlock.
- **Consumer Step:**
 - Check if items exist.
 - Lock the system, remove one full item, increase the empty slot count, and then unlock.
- **Safety Check:**
 - If **e=0**, production is blocked (Full).
 - If **f=0**, consumption is blocked (Empty).
- **Repeat:** Keep the process running in a loop until the user exits.

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int m=1,f=0,e=5,x=0;

int wait(int s){ return --s; }
int signal(int s){ return ++s; }

void p(){
    m=wait(m); e=wait(e); f=signal(f);
    printf("\nProduced %d", ++x);
```

```
    m=signal(m);  
}
```

```
void c(){  
    m=wait(m); f=wait(f); e=signal(e);  
    printf("\nConsumed %d", x--);  
    m=signal(m);  
}
```

```
int main(){  
    int ch;  
    while(1){  
        printf("\n1.P 2.C 3.E: ");  
        scanf("%d",&ch);  
        if(ch==1){  
            if(m==1 && e!=0) p();  
            else printf("\nFull");  
        }  
        else if(ch==2){  
            if(m==1 && f!=0) c();  
            else printf("\nEmpty");  
        }  
        else exit(0);  
    }  
}
```

OUTPUT:

```
[navdeep@localhost os_exp]$ gcc pc.c -o pc
[navdeep@localhost os_exp]$ ./pc
Produced: 0
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[navdeep@localhost os_exp]$
```

RESULT:

The program successfully simulates process synchronization. The producer is restricted from adding items when the buffer is full ($e=0$), and the consumer is restricted from removing items when the buffer is empty ($f=0$), maintaining data integrity within the shared resource.